

# Kali Linux USB Setup Guide with Persistence and VM Support

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## Prerequisites

### Hardware Requirements

- **USB Drive:** Minimum 32GB (64GB+ recommended for better performance)
- **USB 3.0 or higher:** For optimal performance
- **Host Computer:** 4GB+ RAM, 2GB+ free disk space
- **Target Systems:** UEFI and Legacy BIOS support

### Software Requirements

- **Kali Linux ISO:** Latest version from official Kali website
- **Rufus** (Windows) or **Etcher** (Cross-platform) or **dd** (Linux/macOS)
- **GParted** or similar partition manager
- **VirtualBox** or **VMware** (for VM testing)

### Downloads

1. **Kali Linux ISO:** Download from <https://www.kali.org/get-kali/>
  2. **Rufus:** Download from <https://rufus.ie/> (Windows users)
  3. **Etcher:** Download from <https://www.balena.io/etcher/> (Cross-platform)
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## Creating Kali USB with Persistence

### Step 1: Prepare the USB Drive

**Warning:** This process will erase all data on the USB drive. Back up important files first.

### For Windows Users (Using Rufus):

1. Insert your USB drive
2. Launch Rufus as Administrator
3. Configure Rufus settings:
  - **Device:** Select your USB drive
  - **Boot selection:** Click "SELECT" and choose Kali ISO
  - **Partition scheme:** GPT (for UEFI) or MBR (for Legacy BIOS)
  - **Target system:** UEFI (non CSM) or BIOS/UEFI
  - **File system:** FAT32
  - **Cluster size:** 4096 bytes (default)
4. **Enable Persistence:**
  - Check "Show advanced drive properties"
  - Set "Persistent partition size" to desired amount (8GB-16GB recommended)
  - Leave some space for additional partitions
5. Click "START" and wait for completion

### For Linux Users (Using dd and manual partitioning):

#### 1. Identify USB device:

```
bash
```

```
lsblk
```

```
# Example output: /dev/sdb (your USB drive)
```

#### 2. Unmount the USB drive:

```
bash
```

```
sudo umount /dev/sdb*
```

#### 3. Write ISO to USB:

```
bash
```

```
sudo dd if=kali-linux-2024.x-live-amd64.iso of=/dev/sdb bs=4M status=progress
```

```
sudo sync
```

#### 4. Create persistence partition:

```
bash
```

```
# Install parted if not available
```

```
sudo apt install parted
```

```
# Create new partition for persistence
```

```
sudo parted /dev/sdb
```

```
(parted) print free
```

```
(parted) mkpart primary ext4 4GB 100%
```

```
(parted) quit
```

```
# Format persistence partition
```

```
sudo mkfs.ext4 -L persistence /dev/sdb3
```

## 5. Configure persistence:

```
bash
```

```
# Mount the persistence partition
```

```
sudo mkdir -p /mnt/persistence
```

```
sudo mount /dev/sdb3 /mnt/persistence
```

```
# Create persistence configuration
```

```
echo "/ union" | sudo tee /mnt/persistence/persistence.conf
```

```
# Unmount
```

```
sudo umount /dev/sdb3
```

## Step 2: Test the USB Boot

### 1. Boot from USB:

- Restart computer and access boot menu (F12, F8, or DEL key)
- Select USB drive from boot options
- Choose "Live system (persistence)" from Kali boot menu

### 2. Verify Persistence:

- Create a test file on desktop
- Reboot and check if file persists
- Run: `df -h` to verify persistence partition is mounted

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## Setting Up VM Support

## Step 3: Enable Virtualization Features

### Configure Hardware Virtualization:

#### 1. BIOS/UEFI Settings:

- Enable Intel VT-x or AMD-V
- Enable VT-d (Intel) or AMD-Vi (AMD)
- Save and exit BIOS

#### 2. Boot into Kali USB:

- Verify virtualization support:

```
bash
```

```
# Check for virtualization support
```

```
grep -E 'vmx|svm' /proc/cpuinfo
```

```
# Install CPU checker
```

```
sudo apt update
```

```
sudo apt install cpu-checker
```

```
kvm-ok
```

## Step 4: Install Virtualization Software

### Option A: VirtualBox Installation

#### 1. Update system:

```
bash
```

```
sudo apt update && sudo apt upgrade -y
```

#### 2. Install VirtualBox:

```
bash
```

```
sudo apt install virtualbox virtualbox-ext-pack
```

#### 3. Add user to vboxusers group:

```
bash
```

```
sudo usermod -a -G vboxusers $USER
```

#### 4. Configure VirtualBox:

- Launch VirtualBox

- Go to File → Preferences → Extensions
- Install VirtualBox Extension Pack if not already installed

## Option B: VMware Installation

1. **Download VMware Workstation Player** (free for personal use)
2. **Install VMware:**

```
bash
```

```
chmod +x VMware-Player-*.bundle
```

```
sudo ./VMware-Player-*.bundle
```

3. **Install VMware tools:**

```
bash
```

```
sudo apt install open-vm-tools open-vm-tools-desktop
```

## Option C: KVM/QEMU Installation

1. **Install KVM and related packages:**

```
bash
```

```
sudo apt install qemu-kvm libvirt-daemon-system libvirt-clients bridge-utils virt-manager
```

2. **Add user to libvirt group:**

```
bash
```

```
sudo usermod -a -G libvirt $USER
```

3. **Start libvirt service:**

```
bash
```

```
sudo systemctl enable libvirtd
```

```
sudo systemctl start libvirtd
```

## Step 5: Configure VM Networking

### For VirtualBox:

1. **Create Host-Only Network:**
  - File → Host Network Manager
  - Create new host-only network

- Configure IP range (e.g., 192.168.56.0/24)

## 2. **Configure VM Network Settings:**

- Adapter 1: NAT (for internet access)
- Adapter 2: Host-only (for host communication)

### **For VMware:**

## 1. **Configure Virtual Networks:**

- Edit → Virtual Network Editor
  - Add VMnet (Host-only) network
  - Configure subnet settings
- 

# **Post-Installation Configuration**

## **Step 6: Essential Tools and Updates**

### 1. **Update Kali repositories:**

```
bash
```

```
sudo apt update && sudo apt upgrade -y
```

### 2. **Install essential tools:**

```
bash
```

```
# Development tools
```

```
sudo apt install build-essential git vim curl wget
```

```
# Network tools
```

```
sudo apt install nmap wireshark tcpdump netcat
```

```
# Additional penetration testing tools
```

```
sudo apt install metasploit-framework burpsuite sqlmap
```

### 3. **Install Docker (for containerized testing):**

```
bash
```

```
sudo apt install docker.io docker-compose
```

```
sudo systemctl enable docker
```

```
sudo usermod -a -G docker $USER
```

## **Step 7: Configure Persistence Settings**

### 1. Create persistent directories:

```
bash

# Create directories that will persist across reboots
mkdir -p ~/persistent-tools
mkdir -p ~/persistent-configs
mkdir -p ~/persistent-data
```

### 2. Configure shell environment:

```
bash

# Edit .bashrc for persistent aliases and functions
echo "# Persistent custom configurations" >> ~/.bashrc
echo "export PATH=$PATH:~/persistent-tools/bin" >> ~/.bashrc
echo "alias ll='ls -la'" >> ~/.bashrc
```

### 3. Set up SSH keys (if needed):

```
bash

ssh-keygen -t ed25519 -C "your_email@example.com"
```

## Step 8: Configure Boot Options

### 1. Edit GRUB configuration (if needed):

```
bash

sudo nano /etc/default/grub
# Add: GRUB_CMDLINE_LINUX_DEFAULT="quiet splash persistence"
sudo update-grub
```

### 2. Create custom boot entry:

```
bash

# Add custom entry to isolinux.cfg (on USB)
# Mount USB boot partition and edit isolinux/isolinux.cfg
```

---

## Troubleshooting

### Common Issues and Solutions

#### USB Boot Issues:

- **Problem:** USB not recognized as bootable

- **Solution:**
  - Ensure USB is properly formatted
  - Check BIOS boot order
  - Try different USB ports
  - Disable Secure Boot in UEFI

### **Persistence Not Working:**

- **Problem:** Changes don't persist after reboot
- **Solution:**
  - Verify persistence partition is properly created
  - Check persistence.conf file exists
  - Ensure persistence boot option is selected
  - Verify sufficient free space on persistence partition

### **VM Performance Issues:**

- **Problem:** Slow VM performance
- **Solution:**
  - Allocate more RAM to VMs
  - Enable hardware acceleration
  - Use SSD for VM storage
  - Adjust VM settings for performance

### **Network Issues in VMs:**

- **Problem:** No network connectivity in VMs
- **Solution:**
  - Check virtual network adapter settings
  - Verify host network configuration
  - Restart network services
  - Check firewall settings

### **Hardware Compatibility:**

- **WiFi Issues:** Install appropriate drivers
- **Graphics Issues:** Install proprietary drivers if needed



- **USB 3.0 Issues:** Try USB 2.0 ports or update drivers
- 

## Advanced Configuration

### Step 9: Custom Tool Installation

#### 1. Install additional repositories:

```
bash
```

```
# Add external repositories (example)
```

```
echo "deb https://http.kali.org/kali kali-rolling main non-free contrib" | sudo tee /etc/apt/sources.list.d/kali.list
```

#### 2. Install custom tools:

```
bash
```

```
# Example: Install additional security tools
```

```
sudo apt install gobuster dirb nikto
```

```
# Install from GitHub
```

```
cd ~/persistent-tools
```

```
git clone https://github.com/SecureAuthCorp/impacket.git
```

```
cd impacket
```

```
pip3 install .
```

### Step 10: Automation Scripts

#### 1. Create setup automation script:

```
bash
```

```
#!/bin/bash
```

```
# ~/persistent-tools/setup.sh
```

```
# Update system
```

```
sudo apt update && sudo apt upgrade -y
```

```
# Install favorite tools
```

```
sudo apt install -y nmap wireshark burpsuite
```

```
# Configure aliases
```

```
echo "alias scan='nmap -sS -O'" >> ~/.bashrc
```

```
# Set up workspace
```

```
mkdir -p ~/workspace/{recon,exploits,reports}
```

## 2. Create backup script:

```
bash

#!/bin/bash

# ~/persistent-tools/backup.sh

# Backup important configurations
tar -czf ~/persistent-data/backup-$(date +%Y%m%d).tar.gz \
    ~/.bashrc \
    ~/.ssh/ \
    ~/persistent-configs/
```

## Step 11: Multi-Boot Configuration

### 1. Create multiple persistence partitions:

```
bash

# Create different persistence partitions for different use cases
# Partition 1: General use (8GB)
# Partition 2: Forensics work (16GB)
# Partition 3: Development (8GB)
```

### 2. Configure boot menu:

```
bash

# Edit syslinux configuration to add multiple boot options
# Each with different persistence partitions
```

---

## Performance Optimization

### USB Performance Tips:

- Use high-quality USB 3.0+ drives
- Enable write caching (if supported)
- Use ext4 filesystem for persistence
- Allocate sufficient persistence space

### VM Performance Tips:

- Allocate 50-75% of host RAM to VMs
- Enable hardware acceleration

- Use host-only networks when possible
- Store VMs on fast storage (SSD)

### **System Performance:**

- Disable unnecessary services
  - Use lightweight desktop environment
  - Configure swap file on persistence partition
  - Regular cleanup of temporary files
- 

## **Security Considerations**

### **USB Security:**

- Encrypt persistence partition if handling sensitive data
- Use strong passwords for user accounts
- Keep system updated
- Be cautious with untrusted networks

### **VM Security:**

- Isolate VMs from host network when testing
  - Use snapshots before risky operations
  - Don't store sensitive data in VMs
  - Regularly update VM software
- 

## **Maintenance and Updates**

### **Regular Maintenance:**

```
bash
```

```
# Weekly maintenance script
```

```
sudo apt update && sudo apt upgrade -y
```

```
sudo apt autoremove -y
```

```
sudo apt autoclean
```

```
# Check disk usage
```

```
df -h
```

```
# Clean temporary files
```

```
sudo rm -rf /tmp/*
```

## Backup Strategy:

- Regular backups of persistence data
  - Export VM configurations
  - Document custom configurations
  - Keep installation media updated
- 

## Conclusion

This guide provides a comprehensive setup for Kali Linux on USB with persistence and VM support. The configuration allows for:

- Portable penetration testing platform
- Persistent data and configurations
- Full virtualization capabilities
- Professional testing environment

Remember to always use these tools ethically and legally, only on systems you own or have explicit permission to test.

For additional help and community support, visit:

- Kali Linux Documentation: <https://www.kali.org/docs/>
- Kali Linux Forums: <https://forums.kali.org/>
- Official Kali Linux Discord/IRC channels